

On Volatility-Managed Portfolios: The Role of Conditional Factor Correlations

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Abstract

Out of the nine volatility-managed equity factors employed by [Moreira and Muir \(2017\)](#), only three (momentum, ROE, and betting-against-beta) produce robust significantly positive alphas over the 1967–2024 period. However, the evidence for portfolios of factors is considerably stronger. Concretely, the scaled mean-variance-efficient combination of the five factors of [Fama and French \(2015\)](#) produces significantly positive alphas, even when imposing no leverage. The explanation for these apparent contradictory results lies on the important role played by conditional correlations among the individual factors. Our results suggest that the profitability of volatility-management is best judged when incorporating the diversification benefits associated with factors.

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1 Introduction

Building on the work of [Fleming et al. \(2001\)](#), [Marquering and Verbeek \(2004\)](#), and [Barroso and Santa-Clara \(2015\)](#), [Moreira and Muir \(2017\)](#) (MM henceforth) document that the volatility-managed version of several equity factors produces a significantly enhanced risk-return tradeoff relative to the original factors. This can be inferred from the significantly positive alphas obtained from spanning regressions of the scaled factors onto the original counterparts (or alternatively from multiple spanning regressions that also include the three factors of [Fama and French \(1993\)](#)). MM find a similar pattern for various mean-variance-efficient (MVE) combinations of those equity factors, where the scaled MVE portfolios are regressed on the original counterparts.

We reassess the empirical evidence provided in MM over an updated sample period, 1967 to 2024, but relying on the same nine equity factors analyzed in their study. In addition to the one-month realized variances covered in MM, we also consider scaled factors based on three- and six-month realized variances. Our results indicate a clear dichotomy. First, at the individual-factor level, only three risk-managed factors—momentum, net profitability, and betting-against-beta—produce significantly positive alphas. This is especially the case when we impose a Bonferroni adjustment (to account for the multiple-testing problem) and when we exclude the possibility of levered positions (i.e., the weights scaling the original factors are capped at one). This represents a third of the nine equity strategies covered in MM, thus suggesting that the previous evidence on the profitability of the volatility-managed strategies of individual factors is probably overstated.

Second, by defining MVE portfolios of the factors along the lines of MM, we largely confirm their qualitative evidence on the improved risk-return tradeoff associated with volatility-management. In particular, the alphas from spanning regressions of the scaled MVE portfolios on the raw counterparts are positive and strongly significant in nearly all cases, even when we rule out levered positions. Concretely, a notable result holds for the MVE portfolio associated with the five factors of [Fama and French \(2015\)](#) (FF5): The alphas of the scaled FF5 MVE portfolio are always significantly positive, while the same does not occur for any of the corresponding individual factors.

This decoupling suggests that the realized covariances (or correlations) among the factors in a given MVE portfolio have a strong effect in terms of driving the large profitability of the volatility-managed portfolios. Our empirical results largely confirm such a conjecture: Either by setting the conditional correlations (among the factors) to zero or by imposing the unconditional counterparts—which means discarding time-variation in factor correlations in both settings—we find that the profitability of the scaled FF5 MVE portfolio declines

sharply. Specifically, the alphas from the spanning regressions register a substantial decline in magnitude and become insignificant in most cases. Moreover, this pattern holds irrespective of whether we preclude or not levered positions. Our qualitative results survive a battery of robustness checks, namely using several alternative empirical proxies for conditional risk, using a different scaling in the construction of the volatility-managed portfolios, relaying on the idiosyncratic component of realized variance, and employing different sample periods.

Our main takeaway is that time-varying correlations play a major role in terms of driving the profitability of the risk-managed versions of MVE portfolios of factors. Therefore, the performance of volatility-management is best judged when incorporating the diversification benefits associated with the various equity factors from the asset pricing literature. To the best of our knowledge, this pivotal role of factor correlations has been neglected in previous studies on volatility management.

The remainder of the paper proceeds as follows. Section 2 provides the empirical analysis for individual equity factors, while Section 3 contains the corresponding analysis for portfolios of factors. A sensitivity analysis is provided in Section 4. Finally, Section 5 concludes.

2 Volatility-Managed Factors

This section contains the empirical analysis of risk-managed individual factors.

2.1 Empirical Background

Following [Moreira and Muir \(2017\)](#) (MM henceforth), our baseline measure of realized risk for a given equity factor (or long-short portfolio) f is the respective realized variance,¹

$$RV_{f,t}^2 \equiv \frac{21}{K} \sum_{n=0}^{K-1} (f_{d_t-n} - \hat{\mu}_{f,t})^2, \quad (1)$$

where $\hat{\mu}_{f,t} \equiv \frac{1}{K} \sum_{n=0}^{K-1} f_{d_t-n}$ is the average daily factor return in month $t = 1, \dots, T$. f_{d_t-n} denotes a one-day realization of factor f occurring at time $d_t - n$ and $\{d_t\}_{t=1}^T$ represents the time-series of the dates of the last trading session of each month. In addition to the baseline one-month realized variance used by MM, we consider three- and six-month realized variances, that is, $K = 21, 63, 126$, following [Ghysels, Santa-Clara, and Valkanov \(2005\)](#) and [Barroso and Santa-Clara \(2015\)](#).² These longer windows relax the restriction that more

¹In this paper, we do not take a stand on whether a given factor f is a “risk factor”, i.e., a source of systematic risk.

²Our choice of 21 daily returns, rather than 22 daily returns used by MM, follows [Barroso and Santa-Clara \(2015\)](#) and stems from the fact that several months do not have 22 trading sessions. However, this choice

distant (in the past) daily returns do not affect the (empirical) conditional second moments. Hence, this allows us to verify whether our qualitative results are sensitive to the one-month window assumption in the construction of the realized variances.³

As in MM, the risk-managed factor based on $RV_{f,t}^2$ is computed as

$$f_{t+1}^\sigma = \frac{c}{RV_{f,t}^2} f_{t+1}, \quad (2)$$

where c is a constant that equates the sample standard deviations of the managed and raw monthly factors:

$$c = \frac{sd(f_{t+1})}{sd(f_{t+1}/RV_{f,t}^2)}. \quad (3)$$

We consider all nine equity factors covered in MM by relying on a common sample period.⁴ Using a common sample for all factors represents an obvious empirical choice, as it enables a direct comparison of the performance among the different factors, as discussed below.⁵ Moreover, by construction, the volatility-managed portfolios of factors, which are analyzed in the next section, are only feasible under the same time-series period. We include the three factors in FF3 (Mkt , SMB , HML), the momentum factor (Mom) of Carhart (1997), the operating profitability (RMW) and investment (CMA) factors of Fama and French (2015), as well as the profitability (ROE) and investment (IA) counterparts from Hou, Xue, and Zhang (2015). Following MM, we also include the betting-against-beta portfolio (BAB) of Frazzini and Pedersen (2014). We employ the 1967:2–2024:12 sample, where the starting date is defined by the availability of return data for both ROE and IA .⁶ The daily and monthly return data for both ROE and IA are from Lu Zhang’s data library, whereas the corresponding data for BAB are retrieved from AQR’s web site. The daily and monthly return data on the remaining factors are from Kenneth French’s web page.

Table 1 (Panel A) displays the descriptive statistics for the monthly factor returns. It also reports the results from simple AR(1) regressions applied to the realized factor variances:

$$RV_{f,t+1}^2 = \delta + \phi RV_{f,t}^2 + u_{f,t+1}. \quad (4)$$

We can see that the realized variances are persistent across most factors, as the estimates

has a very residual impact in the results reported below.

³We use interchangeably the designations “volatility-managed”, “risk-managed”, or “variance-managed”. Unless otherwise noted, the measure of conditional risk is the realized variance presented above.

⁴Thus, we ignore the currency factor used in MM (FX), as this is not included in the analysis of portfolios of factors.

⁵Such a comparison is not possible under MM’s setup, as they employ different sample periods for each individual factor.

⁶The daily return data between 1967:1 and 2024:11 are used to construct the risk variables.

for ϕ are generally above 0.50. The exceptions hold for both *Mkt* and *SMB*, albeit the respective slope estimates still being strongly statistically significant (1% level).

The summary statistics presented in Panel B of Table 1 show that the Sharpe ratios associated with the MVE portfolios are considerably above the counterparts of the corresponding individual factors. One interesting feature is that the realized variances of *C4* are considerably less persistent than the corresponding second moments for the remaining MVE portfolios.

2.2 Spanning Regressions

Following the related literature, we conduct OLS spanning regressions with monthly returns for each scaled factor (f^σ) using either the original factor (f) or the 3-factor model of Fama and French (1993) augmented with the raw factor (F+FF3) as benchmark models,

$$f_t^\sigma = \alpha + \beta f_t + \varepsilon_t, \quad (5)$$

$$f_t^\sigma = \alpha + \beta f_t + \beta_{Mkt} Mkt_t + \beta_{SMB} SMB_t + \beta_{HML} HML_t + \varepsilon_t, \quad (6)$$

where the statistical significance of the intercept and slope estimates is based on heteroskedasticity-robust standard errors (White (1980)) with the finite-sample correction (i.e., HC1).⁷

The annualized alpha estimates associated with the nine one-month scaled factor strategies are reported in Panel A of Table 2. We can see that the alpha estimates for both Mkt^σ and HML^σ are economically small and statistically insignificant (even at the 10% level), with t -ratios around one. The alphas for the size variance-managed strategy are likewise small but negative and also far from significant.⁸ Among the other equity portfolios, it turns out that the scaled *CMA* produces insignificant alphas (even at the 10% level), while the alphas associated with the other managed investment factor (*IA*) are significant (albeit marginally so when we use F+FF3 as a benchmark model, with a t -ratio of 2.05).

There are some important differences relative to the results provided in MM for a sample period that ends in 2015, namely the finding that the scaled market and value factors are no longer significantly profitable (over the updated sample period). Results provided in the Internet Appendix (IA) Table A.1 show that this is not a feature of the post-2015 period, as estimating the spanning regressions over the 1967–2015 period (i.e., using MM’s ending month) provides similar qualitative results for both Mkt^σ and HML^σ . The implication is

⁷Naturally, in the cases of Mkt^σ , SMB^σ , and HML^σ , the reference model coincides with FF3.

⁸Note that a significantly negative alpha would mean that the managed portfolio provides negative diversification benefits to a portfolio that includes the factors in FF3. That is, a mean-variance investor with access to the other factors can further improve her Sharpe ratio by shorting the managed portfolio.

that the volatility-managed market and value strategies are not relevant in the post-1967 sample.⁹ In opposite direction, our results in the IA (for the pre-2015 period) show that the alpha for IA^σ is not significant when we consider F+FF3 as the reference model.¹⁰ In other words, the profitability of the volatility-managed IA has improved in recent years.

The results discussed above are qualitatively equivalent if we use instead the three- or six-month scaled strategies, in which the starting date is either 1967:4 or 1967:7. The main difference is that the alphas associated with RMW^σ (in the single-factor spanning regression) decrease in magnitude (from 2.76% to either 1.93% or 1.98%) and become significant only at the 5% level. We also observe a relevant decrease in the magnitudes of the alphas for several of the remaining factors, albeit without altering the qualitative statistical significance. Moreover, the scaled market alphas become negative in most cases, albeit with no statistical significance.

Next, we re-run the spanning regressions by imposing no leverage in the definition of the risk-managed factors. This means that whenever the weight (underlying a given managed factor) in a given month is above one, we cap it to one (i.e., $c/RV_{f,t}^2 \leq 1$). The objective is to assess to which extent the results above are driven by extreme weights that indicate excessive leverage and large transaction costs.¹¹ The results for the scaled factors based on one-month realized variances are displayed in Panel B of Table 2. Not surprisingly, we observe a decline in the magnitudes of the alphas (to less than half of the original estimates) for several risk-managed factors, including Mom^σ , RMW^σ , IA^σ , ROE^σ , and BAB^σ . Moreover, the alpha associated with IA^σ ceases to be significant (at the 5% level) when we include the FF3 factors in the spanning regression.

The impact of leverage is even clearer when we consider the scaled factors based on the three- and six-month realized variances. Specifically, the four alphas associated with RMW^σ are no longer significant (even at the 10% level for most of these estimates). These results indicate that a large fraction of the profitability of the variance-managed strategies, as documented above, arises from excessive leverage.

Another aspect that relates to the results above is the multiple test problem: Since we inspect the individual statistical significance of the alpha estimates for nine variance-managed factors, it makes sense to impose a Bonferroni correction when making inference. Thus, given a raw significance level of 5%, the correct significance level becomes $0.56\% = 5\%/9$, thus

⁹A more careful look into MM's Table 1 shows that the alpha for HML^σ (based on the one-factor spanning regression) is actually insignificant at the 5% level, with a t -ratio of $1.97/1.02 = 1.93$. However, the corresponding FF3's alpha is strongly significant (t -ratio of $2.66/1.02 = 2.61$).

¹⁰This is aligned with the evidence provided in MM.

¹¹All the factors are zero-cost by construction. However, $c/RV_{f,t}^2 > 1$ implies a large amount of short-selling (in the short leg of the factor; e.g., the portfolio of past losers), which might be difficult and costly to implement in practice.

implying a critical value of 2.77 for a two-sided test. If we impose such a cut-off on the t -ratios reported in Panel A of Table 2 (i.e., allowing for unconstrained leverage), it follows that two (of the six) alpha estimates associated with RMW^σ (i.e., when considering the single-factor spanning regression and the longer-window realized moments) cease to be significant.

Similarly, if we impose no leverage, it follows that none of the six alphas for RMW^σ is significant (including those based on one-month realized variances), while only half of the alphas associated with IA^σ remain significant.¹² If we impose a raw significance level of 1%, then the situation is even more adverse. This provides another perspective on the relatively weak empirical support for the scaled operating profitability and investment factors.

Therefore, the results above indicate that only three risk-managed strategies— Mom , ROE , and BAB —produce significantly positive alphas over the recent sample period, especially when we exclude the possibility of levered positions. This represents a third of the nine equity strategies studied in MM, thus suggesting that the previous evidence on the relevance of volatility-managed strategies of individual factors is overstated.¹³ In addition, there are substantial concerns about whether the original BAB is a tradable portfolio in the first place (e.g., [Novy-Marx and Velikov \(2022\)](#)), a problem that is likely to be exacerbated for the corresponding scaled portfolio.¹⁴

3 Portfolios of Factors

This section contains the empirical analysis of risk-managed portfolios of factors.

Following MM, we evaluate the volatility-managed versions of mean-variance efficient (MVE) portfolios of the factors used in the previous section. The portfolios are associated with the factors in FF3, FF5 (which coincides with FF3 augmented by RMW and CMA), FF6 (FF5 plus Mom), and HXZ (which stands for the four factors used by [Hou, Xue, and Zhang \(2015\)](#), namely Mkt , IA , ROE , and a size factor).¹⁵ In comparison to MM, we do not consider a portfolio of HXZ+ Mom , as there is a substantial overlapping between the momentum factor and ROE . This choice has also the advantage of matching every portfolio of factors to a multifactor model that has been used to explain the cross-section of expected

¹²Even the remaining three cases show only marginal significance, as the t -ratios varying between 2.79 and 2.86 are very close to the cut-off of 2.77.

¹³We note that the profitability of volatility-managed momentum has been established in [Barroso and Santa-Clara \(2015\)](#). Given that profitability, similar gains for ROE are somehow expected since [Novy-Marx \(2015\)](#) finds that ROE mostly reflects earnings surprises which are strongly associated with momentum.

¹⁴[Barroso, Detzel, and Maio \(2025\)](#) show that the volatility-managed version of a more robust low-beta portfolio produces economically significant gains.

¹⁵The size factor employed by [Hou, Xue, and Zhang \(2015\)](#) (ME) differs from SMB , albeit the correlation is very large.

stock returns.¹⁶

The portfolio weights are computed over the full sample period (i.e., 1967:2–2024:12) using

$$\mathbf{w}_{mve} = \frac{\boldsymbol{\Sigma}_f^{-1} \boldsymbol{\mu}_f}{\mathbf{1}^\top \boldsymbol{\Sigma}_f^{-1} \boldsymbol{\mu}_f} \quad (7)$$

where $\mathbf{w}_{mve}$ denotes the mean-variance efficient portfolio weight vector, $\boldsymbol{\Sigma}_f$ is the covariance matrix of monthly factor returns, $\boldsymbol{\mu}_f$ is the vector of mean monthly factor returns, and $\mathbf{1}$ is a vector of ones.

By combining these portfolio weights with the monthly and daily factor returns, we construct monthly and daily MVE portfolio returns. The realized variances of the MVE portfolios are constructed in the same way as for the raw factors. Similarly to the individual factors, the volatility-managed return for MVE portfolio p is given by

$$p_{t+1}^\sigma = \frac{c}{RV_{p,t}^2} p_{t+1}, \quad (8)$$

where $c = sd(p_{t+1})/sd(p_{t+1}/RV_{p,t}^2)$. We run spanning single-regressions of each scaled portfolio return on the original MVE portfolio return:

$$p_t^\sigma = \alpha + \beta p_t + \varepsilon_t. \quad (9)$$

The results are displayed in Table 3 (Panel A). Starting with the strategies based on the one-month realized variances, we see that all five scaled MVE portfolios produce positive alphas that are strongly significant (1% level). Perhaps unsurprisingly, the MVE portfolios with stronger alphas are those including momentum or the related *ROE* factor. But $FF3^\sigma$ (alpha of 2.28%) and $FF5^\sigma$ (alpha of 1.23%) also show strongly significant gains, something that appears in clear contrast with the results displayed in Panel A of Table 2: with the exception of *RMW*, all factors in these two models generate insignificant alphas in the corresponding spanning regressions. Yet, somehow, the *combination* of those same factors already benefits from volatility management.

When we consider the strategies based on the three- and six-month realized variances, we see that the alphas associated with $FF3^\sigma$ have about half the magnitude and become insignificant (even at the 10% level). In comparison, the alphas associated with $FF5^\sigma$ (as well as the respective *t*-ratios) barely change from the counterparts based on the one-month realized variance.

The inconsistency between $FF5^\sigma$ and the scaled versions of the corresponding individual

¹⁶Specifically, FF6 has been used in [Fama and French \(2018\)](#).

factors becomes even more evident when we discard the possibility of levered positions (i.e. $c/RV_{p,t}^2$ is always below one). The results are presented in Table 3 (Panel B). Not surprisingly, the alphas register a substantial decline in magnitude relative to the counterparts reported in Panel A of Table 3. However, the estimates associated with $FF5^\sigma$ are statistically significant (5% level) irrespective of the version of realized variance being used. This marks a clear contrast with the results reported in Panel B of Table 2, where only RMW (and based exclusively on the one-month realized variance) produces a significant alpha (which becomes nonetheless insignificant by imposing the more conservative Bonferroni inference).¹⁷

These results suggest that the realized covariances (or correlations) among the factors in a given portfolio have a strong effect in driving the large profitability of the volatility-managed portfolios of factors, particularly the case of $FF5^\sigma$. To check for this possibility, we compute two alternative versions of the realized variance of each MVE portfolio. The original realized variances of the MVE portfolios ($RV_{p,t}^2$) can be written as

$$RV_{p,t}^2 \equiv \mathbf{w}_{mve}^\top \boldsymbol{\Sigma}_{f,t} \mathbf{w}_{mve} = \mathbf{w}_{mve}^\top \mathbf{D}_{rv,t} \mathbf{C}_{f,t} \mathbf{D}_{rv,t} \mathbf{w}_{mve}, \quad (10)$$

where $\boldsymbol{\Sigma}_{f,t}$ is the realized covariance matrix of underlying factor returns, $\mathbf{C}_{f,t}$ is the realized correlation matrix, and $\mathbf{D}_{rv,t}$ is the diagonal matrix of realized volatilities of underlying factors.

First, we ignore the realized covariance among the factors when computing the portfolio's realized variance (which is equivalent to imposing zero realized correlations), i.e., we replace $\mathbf{C}_{f,t}$ in Equation (10) with an identity matrix.

Taking $FF3$ as an example, the corresponding portfolio's variance is now given by

$$RV_{FF3,t}^2 = \omega_{Mkt}^2 RV_{Mkt,t}^2 + \omega_{SMB}^2 RV_{SMB,t}^2 + \omega_{HML}^2 RV_{HML,t}^2. \quad (11)$$

Second, we replace the realized correlations among the factors by the unconditional counterparts, i.e., we replace $\mathbf{C}_{f,t}$ in Equation (10) with $\mathbf{C}_f$, which is calculated over the full sample period using daily returns. In the case of $FF3$, the realized variance is now given by

$$\begin{aligned} RV_{FF3,t}^2 = & \omega_{Mkt}^2 RV_{Mkt,t}^2 + \omega_{SMB}^2 RV_{SMB,t}^2 + \omega_{HML}^2 RV_{HML,t}^2 + 2\omega_{Mkt}\omega_{SMB} RC_{Mkt,SMB,t} + \\ & 2\omega_{Mkt}\omega_{HML} RC_{Mkt,HML,t} + 2\omega_{SMB}\omega_{HML} RC_{SMB,HML,t}, \end{aligned} \quad (12)$$

where $RC_{j,k,t}$ denotes the realized covariance between factors j and k at time t . This empirical conditional moment is computed by assuming that the conditional correlation is constant

¹⁷Note that the Bonferroni adjustment is less justified in the case of the MVE portfolios (compared to the individual factors), since the five portfolios largely overlap in the factors they contain.

over time and coincides with the respective unconditional correlation,

$$RC_{j,k,t} = \rho_{j,k} RV_{j,t} RV_{k,t}, \quad (13)$$

where $\rho_{j,k}$ denotes the sample correlation between factors j and k (based on daily returns over the entire sample period). In this setting, the conditional factor covariances are allowed to vary over time, but such variation stems exclusively from time variation in the individual conditional variances.

The results displayed in Panel A of Table 4 (unconstrained leverage) show that ignoring time-varying correlations (covariances) among the factors has a tremendous negative effect on the alpha estimates. Specifically, most of the alphas decline to values that are nearly half the original estimates reported in Panel A of Table 3. Interestingly, the alphas associated with both cases (disregarding correlations or using unconditional correlations) are very similar across the board, which suggests that imposing zero or unconditional correlations makes little difference for the profitability of the volatility-managed portfolio strategies.

Results for $FF5^\sigma$ are particularly meaningful for two reasons. First, this set of factors does not include momentum-related factors. Second, it consists of factors with more persistent signals which are shown to be more relevant describing the feasible investment opportunity set of realistic investors exposed to frictions (Detzel et al., 2023). The results for this model show that, shutting down the correlations channel, alphas decline to less than half in the case of the scaled strategy based on one-month realized moments and to around or less than a third the original estimates for the strategies based on three- and six-month realized variances. Importantly, all six alphas associated with $FF5^\sigma$ are clearly not significant, even before considering multiple hypothesis adjustments. ¹⁸

By imposing no leverage, the results in Panel B of Table 4 show a similar qualitative picture. The key difference is that the decline in alphas (by shutting down the time-varying-correlation channel) is smaller than in the baseline case with unrestricted leverage. Yet, the alphas associated with $FF5^\sigma$ register very modest magnitudes, varying between 0.22% and 0.40% (when disregarding correlations) and between 0.21% and 0.39% (when imposing unconditional correlations). Moreover, with the exception of the estimates associated with the one-month realized moments, all alphas for $FF5^\sigma$ are insignificant (even at the 10% level). These results are clearly at odds with the evidence for the baseline setting in Panel B of Table 3.

Overall, the results from this section show that time-varying correlations play a major

¹⁸The contrast of this result with that of $FF6^\sigma$, where all six alphas are strongly significant (at the 1% level) is telling. The only difference between the two models is the inclusion of the momentum factor in the second. Hence, the effectiveness of volatility management for $FF6^\sigma$ comes from momentum.

role in driving the profitability of the risk-managed versions of MVE portfolios of factors.

4 Sensitivity Analysis

In this section, we conduct a sensitivity analysis to the results in the previous section. To keep the focus and save space, we restrict the analysis in this section to $FF5^\sigma$.

First, to assess the sensitivity of the results to RV_p^2 , we use two alternative metrics of realized risk. The first measure is the realized second moment,

$$RSM_{p,t} \equiv \frac{21}{K} \sum_{n=0}^{K-1} p_{dt-n}^2, \quad (14)$$

a metric that is widely used in the risk-return tradeoff literature given that $\hat{\mu}_{p,t}$ is likely a noisier estimate of expected returns at a daily frequency than a constant set to zero. The second measure is the realized volatility,

$$RV_{p,t} \equiv \sqrt{RSM_{p,t}}, \quad (15)$$

which is also widely used in the related literature (e.g., Barroso and Detzel (2021), Barroso and Maio (2024), and Barroso, Detzel, and Maio (2025)). The main advantage of RV is that it is less subjective to spikes than the other two measures (e.g., Barroso and Detzel (2021)).

Second, and following MM, we employ the expected variance ($\mathbb{E}_t(RV_{p,t+1}^2)$), rather than the realized variance ($RV_{p,t}^2$), in the definition of the scaled portfolio. The expected variance is defined as the fitted value of a predictive regression for RV_p^2 ,

$$\mathbb{E}_t(RV_{p,t+1}^2) = \hat{\delta} + \hat{\phi}RV_{p,t}^2, \quad (16)$$

with a hat denoting the corresponding OLS estimate; the predictor, $RV_{p,t}^2$, is computed using a 1-, 3-, or 6-month estimation window.

Third, in line with Barroso and Santa-Clara (2015), where c is common across all portfolios, we use a constant risk target by setting c equal to the variance of the unadjusted MVE portfolio over the full sample period. The choice of c can affect the alpha estimates, albeit it has no impact on the respective statistical significance.

Fourth, we construct alternative volatility-managed portfolios. Specifically, we consider only the idiosyncratic component of the portfolio realized variance, by relying on the CAPM as a benchmark. Based on the following CAPM regression (applied to daily returns and

using the full sample) for a given MVE portfolio p ,

$$p_n = \alpha + \beta_p Mkt_n + \varepsilon_n, n = 1, \dots, T_n, \quad (17)$$

the idiosyncratic realized variance is defined as $RV_{\varepsilon,t}^2$, where

$$RV_{\varepsilon,t}^2 \equiv \frac{21}{K} \sum_{n=0}^{K-1} (\varepsilon_{d_t-n} - \hat{\mu}_{\varepsilon,t})^2, \quad (18)$$

and $\hat{\mu}_{\varepsilon,t} \equiv (1/K) \sum_{n=0}^{K-1} \varepsilon_{d_t-n}$.

This is equivalent to the realized variance of the MVE portfolio orthogonal to the market factor:

$$RV_{\varepsilon,t}^2 \equiv \tilde{\mathbf{w}}^\top \Sigma_{f,t} \tilde{\mathbf{w}} = \tilde{\mathbf{w}}^\top \mathbf{D}_{rv,t} \mathbf{C}_{f,t} \mathbf{D}_{rv,t} \tilde{\mathbf{w}}, \quad (19)$$

where $\tilde{\mathbf{w}}$ is obtained from $\mathbf{w}_{mve}$ by subtracting β_p from the market-factor weight, while keeping the weights of all other factors unchanged.

Fifth, we employ a sample period that begins in 1963 and ends in 2015. This coincides with MM's original sample period for the MVE portfolio of FF5 factors and facilitates the comparison between the two studies. It also provides evidence on whether financial markets turbulence in recent years (e.g., the COVID crisis) has an impact on our results.

Lastly, we use a sample period determined by the availability of the FF5 factors, which starts in 1963 and ends in 2024.

The results are provided in Tables A.2. We obtain the same qualitative picture as in the benchmark setting discussed in the last section: The alphas associated with the five-factor model suffer a substantial decline in magnitude and become statistically insignificant when we rule out time-varying correlations among the factors. Moreover, this pattern is robust to whether we impose zero correlations or work with unconditional correlations. Such a finding is also robust to whether we work with the original unconstrained strategies or impose zero leverage in their construction.

5 Conclusion

Building on previous literature, [Moreira and Muir \(2017\)](#) uncover a recurrent pattern in financial markets whereby factor-level volatility can be consistently used to profitably time a broad set of leading equity factors. Our findings offer a reappraisal of these conclusions. In a unified setting, controlling for multiple hypothesis testing, and with an extended and updated sample, the evidence for the effectiveness of timing volatility of individual factors turns out

to be more circumscribed. The case for timing is fragile even before the pertinent estimation issues pointed out by [Cederburg et al. \(2020\)](#), let alone the trading cost implications studied in [Barroso and Detzel \(2021\)](#) and in [DeMiguel, Martín-Utrera, and Uppal \(2024\)](#). Focusing only on gross alphas, the most benevolent possible setting for volatility management, the null of zero alpha is simply not rejected for six factors out of nine. The benefits do stand out for momentum, the related *ROE* factor, and for betting-against-beta. Overall, these findings shift the status of the literature away from asking why volatility in general might be mispriced in the time series. It rather prompts the question of why some specific factors would gain so strongly from timing volatility in the first place.

Yet, going beyond individual factors, to multifactor portfolios, the big picture of [Moreira and Muir \(2017\)](#) comes back strongly. Even FF5, a model where no single factor gains from volatility timing, produces an MVE portfolio that does benefit from timing. However, factor volatilities per se do not seem to account for this pattern. Decomposing the risk of the MVE portfolio into variances and factor correlations, correlations turn out to play an essential role. Specifically, the alpha of the volatility-managed portfolio comes from reducing exposure to the MVE whenever its underlying factors become more correlated. Shutting down this time-varying factor correlation channel, volatility management achieves mediocre results for this preeminent asset pricing model as well.

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Table 1 Summary Statistics

The table reports the mean, standard deviation, and Sharpe ratio of individual factors (Panel A) and mean–variance efficient portfolios (Panel B), along with AR(1) estimates for their one-month, three-month, and six-month realized variances based on non-overlapping windows. Except for ϕ , all measures are annualized. Except for the Sharpe ratio and ϕ , all measures are expressed in percentage terms. The sample period is 1967:2–2024:12. The three-month (six-month) realized variances start in 1967:5 (1967:7). * and ** denote statistical significance at the 5% and 1% level, respectively.

				AR(1): 1M RV ²		AR(1): 3M RV ²		AR(1): 6M RV ²	
	Mean	Std	SR	δ	ϕ	δ	ϕ	δ	ϕ
Panel A: Individual Factors									
Mkt	7.07	15.80	0.45	1.49** (7.20)	0.42** (12.34)	1.65** (5.39)	0.39** (6.49)	1.99** (4.79)	0.27** (2.99)
SMB	1.61	10.58	0.15	0.50** (9.58)	0.32** (8.83)	0.53** (6.95)	0.32** (5.08)	0.57** (5.78)	0.26** (2.90)
HML	3.27	10.52	0.31	0.23** (4.86)	0.73** (27.71)	0.22** (2.94)	0.75** (17.20)	0.33** (2.70)	0.63** (8.51)
MOM	7.12	14.77	0.48	0.59** (5.47)	0.60** (19.52)	0.49** (3.15)	0.68** (14.15)	0.70** (2.86)	0.55** (7.03)
RMW	3.57	7.82	0.46	0.11** (5.15)	0.71** (26.35)	0.15** (3.56)	0.65** (12.79)	0.14* (2.54)	0.68** (9.78)
CMA	3.26	7.19	0.45	0.11** (6.66)	0.67** (23.82)	0.10** (3.70)	0.73** (16.11)	0.10** (2.75)	0.73** (11.27)
IA	4.16	7.14	0.58	0.13** (5.92)	0.68** (24.16)	0.12** (3.41)	0.72** (15.55)	0.14** (2.79)	0.67** (9.65)
ROE	6.45	9.03	0.71	0.13** (6.30)	0.71** (26.51)	0.16** (4.13)	0.68** (13.89)	0.18** (3.29)	0.64** (8.78)
BAB	9.93	11.46	0.87	0.31** (4.87)	0.66** (23.05)	0.29** (2.94)	0.70** (14.69)	0.44** (2.84)	0.54** (6.76)
Panel B: Mean-Variance Efficient Portfolios									
FF3	4.71	7.67	0.61	0.21** (4.28)	0.61** (20.41)	0.21** (2.69)	0.65** (12.80)	0.32** (2.64)	0.45** (5.27)
C4	5.40	5.81	0.93	0.19** (7.93)	0.33** (9.04)	0.21** (6.17)	0.28** (4.42)	0.22** (5.16)	0.23* (2.53)
FF5	3.91	3.75	1.04	0.05** (7.26)	0.57** (18.16)	0.05** (4.69)	0.58** (10.86)	0.05** (3.27)	0.63** (8.64)
FF6	4.39	3.66	1.20	0.06** (7.70)	0.48** (14.21)	0.06** (5.00)	0.51** (8.94)	0.05** (3.63)	0.55** (6.89)
HXZ	5.24	4.02	1.30	0.06** (8.34)	0.55** (17.19)	0.07** (5.84)	0.48** (8.18)	0.06** (3.82)	0.59** (7.71)

Table 2 Variance-Managed Factor Alphas

This table presents annualized alphas (in %), and respective heteroskedasticity-robust t-ratios in parentheses, associated with nine variance-managed equity factors. The reference model is either a single-factor model containing the original factor (F) or a multifactor model that contains the original factor plus the factors in the Fama-French three-factor model (F+FF3). Each managed factor scales the raw factor by the inverse of its one-month, three-month, or six-month realized variances. In Panel B, the weights associated with the managed factors are bounded at one. The sample periods are 1967:2–2024:12, 1967:5–2024:12, and 1967:7–2024:12 for one-month, three-month, and six-month realized variances, respectively. * and ** denote statistical significance at the 5% and 1% level, respectively.

	Mkt^σ	SMB^σ	HML^σ	Mom^σ	RMW^σ	CMA^σ	IA^σ	ROE^σ	BAB^σ
Panel A: Unconstrained Leverage									
<i>One-Month Realized Variance</i>									
F	1.52 (1.02)	−1.01 (−1.00)	1.02 (0.93)	10.78** (6.33)	2.76** (3.23)	0.44 (0.65)	1.74** (2.59)	5.25** (5.69)	9.34** (7.91)
F + FF3	1.56 (1.02)	−1.27 (−1.21)	1.12 (0.98)	9.83** (5.86)	3.13** (3.67)	0.09 (0.13)	1.36* (2.05)	5.20** (5.74)	8.67** (7.48)
<i>Three-Month Realized Variance</i>									
F	−0.35 (−0.26)	−1.31 (−1.43)	0.82 (0.81)	11.00** (6.57)	1.93* (2.35)	0.61 (0.95)	1.88** (2.97)	5.34** (5.93)	8.23** (6.82)
F + FF3	−0.32 (−0.23)	−1.65 (−1.74)	0.99 (0.96)	9.80** (5.92)	2.36** (2.87)	0.34 (0.52)	1.62* (2.54)	5.43** (6.12)	7.32** (6.12)
<i>Six-Month Realized Variance</i>									
F	−0.02 (−0.02)	−0.85 (−0.97)	0.56 (0.56)	10.30** (6.22)	1.98* (2.50)	0.83 (1.32)	1.80** (2.90)	5.04** (5.77)	7.49** (6.58)
F + FF3	0.09 (0.07)	−1.13 (−1.26)	0.80 (0.78)	8.96** (5.48)	2.39** (3.04)	0.64 (1.01)	1.64** (2.60)	5.12** (5.96)	6.62** (5.97)
Panel B: No Leverage									
<i>One-Month Realized Variance</i>									
F	0.49 (0.67)	−0.35 (−0.77)	0.89 (1.86)	3.94** (4.95)	0.88* (2.15)	0.15 (0.47)	0.68* (2.13)	2.30** (4.43)	3.71** (5.98)
F + FF3	0.57 (0.75)	−0.56 (−1.17)	1.05* (2.12)	3.59** (4.49)	1.13** (2.72)	−0.04 (−0.11)	0.54 (1.63)	2.42** (4.75)	3.24** (5.13)
<i>Three-Month Realized Variance</i>									
F	−0.20 (−0.26)	−0.58 (−1.34)	0.63 (1.30)	4.13** (5.09)	0.47 (1.16)	0.39 (1.21)	0.92** (2.83)	2.29** (4.43)	3.29** (5.36)
F + FF3	−0.12 (−0.16)	−0.82 (−1.85)	0.82 (1.65)	3.63** (4.42)	0.73 (1.80)	0.27 (0.81)	0.90** (2.61)	2.41** (4.74)	2.82** (4.56)
<i>Six-Month Realized Variance</i>									
F	−0.16 (−0.22)	−0.53 (−1.27)	0.59 (1.20)	3.85** (4.61)	0.39 (1.02)	0.40 (1.25)	0.93** (2.86)	2.20** (4.25)	3.08** (5.09)
F + FF3	−0.05 (−0.07)	−0.73 (−1.75)	0.85 (1.69)	3.18** (3.83)	0.63 (1.64)	0.36 (1.07)	0.95** (2.79)	2.34** (4.59)	2.56** (4.22)

Table 3 Variance-Managed Alphas for Mean-Variance Efficient Portfolios

This table presents annualized alphas (in %), and respective heteroskedasticity-robust t-ratios in parentheses, associated with variance-managed mean-variance efficient (MVE) portfolios. Unconditional MVE portfolios are formed using various combinations of factors. The reference model is a single-factor model containing the unconditional MVE portfolio. Each managed MVE portfolio scales the raw MVE portfolio by the inverse of its one-month, three-month, or six-month realized variances. In Panel B, the weights associated with the managed MVE portfolios are bounded at one. The sample periods are 1967:2–2024:12, 1967:5–2024:12, and 1967:7–2024:12 for one-month, three-month, and six-month realized variances, respectively. * and ** denote statistical significance at the 5% and 1% level, respectively.

	$FF3^\sigma$	$C4^\sigma$	$FF5^\sigma$	$FF6^\sigma$	HXZ^σ
Panel A: Unconstrained Leverage					
1M RV^2	2.28** (2.91)	3.11** (5.20)	1.23** (3.49)	2.09** (5.37)	2.22** (6.02)
3M RV^2	1.05 (1.33)	2.18** (3.67)	1.07** (3.30)	1.95** (5.38)	1.99** (5.75)
6M RV^2	1.16 (1.48)	1.81** (3.13)	1.11** (3.44)	1.83** (5.23)	1.83** (5.63)
Panel B: No Leverage					
1M RV^2	0.71 (1.73)	1.53** (4.97)	0.58** (3.28)	1.05** (5.28)	1.01** (4.63)
3M RV^2	0.52 (1.22)	1.16** (3.54)	0.42* (2.22)	0.87** (4.14)	0.95** (4.23)
6M RV^2	0.67 (1.62)	0.89** (2.72)	0.40* (2.26)	0.79** (3.59)	0.85** (3.95)

Table 4 Variance-Managed Alphas for Mean-Variance Efficient Portfolios: Static Correlations

This table presents annualized alphas (in %), and respective heteroskedasticity-robust t-ratios in parentheses, associated with variance-managed mean-variance efficient (MVE) portfolios. Unconditional MVE portfolios are formed using various combinations of factors. The reference model is a univariate model with the unconditional MVE portfolio on the right hand side. Each managed MVE portfolio scales the raw MVE portfolio by the inverse of its realized variance estimated with one to six months of daily returns. Factor correlations are either disregarded or replaced by full-sample unconditional correlations when computing the realized variance of each MVE portfolio. In Panel B, the weights associated with the managed MVE portfolios are bounded at one. The sample periods are 1967:2–2024:12, 1967:5–2024:12, and 1967:7–2024:12 for one-month, three-month, and six-month realized variances, respectively. * and ** denote statistical significance at the 5% and 1% level, respectively.

	$FF3^\sigma$	$C4^\sigma$	$FF5^\sigma$	$FF6^\sigma$	HXZ^σ
Panel A: Unconstrained Leverage					
<i>Disregarding Correlations</i>					
1M RV^2	1.43 (1.90)	2.31** (3.86)	0.60 (1.72)	1.38** (3.76)	1.41** (3.90)
3M RV^2	0.38 (0.53)	1.63** (2.73)	0.27 (0.77)	1.02** (2.68)	1.28** (3.44)
6M RV^2	0.60 (0.88)	1.64** (2.81)	0.33 (0.97)	1.05** (2.82)	1.13** (3.11)
<i>Unconditional Correlations</i>					
1M RV^2	1.46 (1.94)	2.34** (3.93)	0.58 (1.65)	1.36** (3.66)	1.40** (3.87)
3M RV^2	0.37 (0.52)	1.63** (2.75)	0.26 (0.72)	1.04** (2.68)	1.30** (3.47)
6M RV^2	0.61 (0.89)	1.65** (2.87)	0.31 (0.90)	1.05** (2.76)	1.13** (3.08)
Panel B: No Leverage					
<i>Disregarding Correlations</i>					
1M RV^2	0.67 (1.62)	1.29** (3.61)	0.40* (2.09)	0.77** (3.48)	0.81** (3.38)
3M RV^2	0.46 (1.13)	1.08** (2.86)	0.29 (1.50)	0.67** (2.77)	0.82** (3.36)
6M RV^2	0.39 (0.99)	0.99** (2.66)	0.22 (1.12)	0.60** (2.59)	0.71** (3.02)
<i>Unconditional Correlations</i>					
1M RV^2	0.69 (1.67)	1.31** (3.72)	0.39* (2.01)	0.77** (3.42)	0.81** (3.32)
3M RV^2	0.45 (1.11)	1.07** (2.84)	0.28 (1.43)	0.68** (2.76)	0.82** (3.34)
6M RV^2	0.38 (0.96)	0.98** (2.66)	0.21 (1.07)	0.62** (2.59)	0.71** (2.99)

A Internet Appendix: Additional Results

This appendix contains results that are discussed, but not tabulated, in the main paper.

Table A.1 Variance-Managed Factor Alphas: 1967–2015

This table presents annualized alphas (in %), and respective heteroskedasticity-robust t-ratios in parentheses, associated with nine variance-managed equity factors. The reference model is either a single-factor model containing the original factor (F) or a multifactor model that contains the original factor plus the factors in the Fama-French three-factor model (F+FF3). Each managed factor scales the raw factor by the inverse of its one-month or six-month realized variance. In Panel B, the weights associated with the managed factors are bounded at one. The sample periods are 1967:2–2015:12, 1967:5–2015:12, and 1967:7–2015:12 for one-month, three-month, and six-month realized variances, respectively. * and ** denote statistical significance at the 5% and 1% level, respectively.

	Mkt^σ	SMB^σ	HML^σ	Mom^σ	RMW^σ	CMA^σ	IA^σ	ROE^σ	BAB^σ
Panel A: Unconstrained Leverage									
<i>One-Month Realized Variance</i>									
F	1.68 (1.09)	−1.16 (−1.03)	0.58 (0.54)	11.71** (6.12)	3.12** (3.34)	0.31 (0.46)	1.49* (2.30)	5.04** (5.24)	10.20** (7.56)
F + FF3	1.73 (1.07)	−1.58 (−1.36)	0.46 (0.40)	10.92** (5.76)	3.92** (4.15)	−0.14 (−0.20)	0.91 (1.45)	5.37** (5.54)	9.39** (7.25)
<i>Three-Month Realized Variance</i>									
F	−0.32 (−0.23)	−1.55 (−1.54)	0.37 (0.37)	11.75** (6.22)	2.29* (2.55)	0.50 (0.78)	1.39* (2.33)	5.23** (5.57)	9.02** (6.49)
F + FF3	−0.31 (−0.21)	−2.04 (−1.93)	0.32 (0.31)	10.60** (5.65)	3.19** (3.53)	0.10 (0.15)	0.87 (1.48)	5.68** (6.01)	8.03** (6.00)
<i>Six-Month Realized Variance</i>									
F	−0.13 (−0.09)	−1.01 (−1.04)	0.09 (0.09)	10.91** (5.85)	2.37** (2.74)	0.67 (1.08)	1.21* (2.11)	4.94** (5.44)	8.21** (6.29)
F + FF3	0.01 (0.01)	−1.47 (−1.49)	0.07 (0.07)	9.71** (5.26)	3.26** (3.79)	0.33 (0.53)	0.80 (1.39)	5.38** (5.87)	7.25** (5.82)
Panel B: No Leverage									
<i>One-Month Realized Variance</i>									
F	0.44 (0.56)	−0.41 (−0.79)	0.72 (1.45)	4.43** (4.73)	1.11* (2.37)	0.16 (0.50)	0.71* (2.21)	2.24** (3.90)	4.05** (5.57)
F + FF3	0.51 (0.61)	−0.71 (−1.33)	0.76 (1.49)	4.11** (4.42)	1.61** (3.41)	−0.11 (−0.33)	0.44 (1.33)	2.59** (4.45)	3.45** (4.89)
<i>Three-Month Realized Variance</i>									
F	−0.15 (−0.18)	−0.69 (−1.42)	0.44 (0.87)	4.50** (4.67)	0.67 (1.46)	0.37 (1.12)	0.76* (2.33)	2.34** (4.07)	3.65** (5.03)
F + FF3	−0.02 (−0.02)	−1.00* (−2.01)	0.54 (1.04)	3.95** (4.13)	1.23** (2.69)	0.13 (0.39)	0.54 (1.60)	2.65** (4.58)	3.08** (4.38)
<i>Six-Month Realized Variance</i>									
F	0.06 (0.08)	−0.61 (−1.33)	0.42 (0.83)	4.11** (4.15)	0.60 (1.38)	0.36 (1.09)	0.76* (2.42)	2.25** (3.89)	3.36** (4.70)
F + FF3	0.29 (0.37)	−0.92 (−1.95)	0.54 (1.04)	3.48** (3.57)	1.15** (2.70)	0.20 (0.59)	0.60 (1.84)	2.55** (4.39)	2.74** (3.98)

Table A.2 Sensitivity Analysis: Risk-Managed Alphas for MVE portfolio of FF5 Factors

This table presents sensitivity analysis for the risk-managed MVE portfolio of FF5 factors based on various risk measures and sample periods. The annualized alphas (in %) relative to the unadjusted MVE portfolio and respective heteroskedasticity-robust t-ratios are reported in parentheses. Cond. ρ , Uncond. ρ and $\rho = 0$ respectively denote conditional (unmodified), unconditional, and zero correlations, respectively, used to compute realized variances. * and ** denote statistical significance at the 5% and 1% level, respectively.

	Unconstrained Leverage			No Leverage		
	Cond. ρ	Uncond. ρ	$\rho = 0$	Cond. ρ	UnCond. ρ	$\rho = 0$
Panel A: Alternative Risk Measures						
1M RSM	1.10** (3.16)	0.50 (1.41)	0.52 (1.50)	0.51** (2.84)	0.39* (2.00)	0.40* (2.10)
3M RSM	1.08** (3.36)	0.26 (0.72)	0.27 (0.76)	0.41* (2.19)	0.28 (1.44)	0.29 (1.52)
6M RSM	1.09** (3.41)	0.30 (0.87)	0.32 (0.95)	0.39* (2.21)	0.21 (1.08)	0.22 (1.13)
1M RV	1.10** (3.16)	0.50 (1.41)	0.52 (1.50)	0.51** (2.84)	0.39* (2.00)	0.40* (2.10)
3M RV	1.08** (3.36)	0.26 (0.72)	0.27 (0.76)	0.41* (2.19)	0.28 (1.44)	0.29 (1.52)
6M RV	1.09** (3.41)	0.30 (0.87)	0.32 (0.95)	0.39* (2.21)	0.21 (1.08)	0.22 (1.13)
Expected 1M RV ²						
1M RV ²	0.90** (3.63)	0.46* (1.97)	0.36* (2.14)	0.37** (2.63)	0.32* (2.36)	0.24* (2.54)
3M RV ²	0.80** (3.13)	0.27 (1.06)	0.23 (1.26)	0.31* (2.11)	0.27 (1.91)	0.21* (2.17)
6M RV ²	0.84** (3.23)	0.27 (1.07)	0.22 (1.26)	0.28 (1.94)	0.23 (1.60)	0.17 (1.84)
Panel B: Constant Risk Target						
1M RV ²	3.00** (3.49)	1.27 (1.65)	0.74 (1.72)	0.38* (2.41)	0.41* (2.49)	0.42* (2.23)
3M RV ²	2.15** (3.30)	0.51 (0.72)	0.30 (0.77)	0.31* (2.07)	0.35* (2.16)	0.31 (1.63)
6M RV ²	2.11** (3.44)	0.57 (0.90)	0.34 (0.97)	0.20 (1.30)	0.29 (1.78)	0.23 (1.19)
Panel C: Realized Variance: Non-Market Component						
1M RV ² _{⊥mkt}	1.05** (2.80)	0.55 (1.51)	0.55 (1.52)	0.75** (3.83)	0.43* (2.15)	0.43* (2.17)
3M RV ² _{⊥mkt}	1.00** (2.84)	0.38 (1.04)	0.34 (0.95)	0.59** (2.80)	0.32 (1.56)	0.31 (1.53)
6M RV ² _{⊥mkt}	1.05** (3.04)	0.37 (1.05)	0.36 (1.04)	0.56** (2.74)	0.22 (1.08)	0.23 (1.14)
Panel D: 1963–2015 Sample						
1M RV ²	1.13** (3.30)	0.65 (1.96)	0.70* (2.11)	0.76** (4.37)	0.51** (2.66)	0.53** (2.80)
3M RV ²	1.20** (3.81)	0.45 (1.33)	0.44 (1.34)	0.58** (2.91)	0.41* (2.06)	0.41* (2.13)
6M RV ²	1.16** (3.74)	0.39 (1.19)	0.41 (1.27)	0.53** (2.88)	0.29 (1.51)	0.31 (1.64)
Panel E: 1963–2024 Sample						
1M RV ²	1.18** (3.48)	0.54 (1.62)	0.56 (1.69)	0.57** (3.38)	0.37* (2.01)	0.38* (2.07)
3M RV ²	1.11** (3.53)	0.27 (0.80)	0.27 (0.81)	0.43* (2.35)	0.27 (1.41)	0.27 (1.46)
6M RV ²	1.09** (3.50)	0.27 (0.82)	0.28 (0.88)	0.40* (2.35)	0.17 (0.90)	0.18 (0.95)